Nicolò Bonacorsi

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Education

Columbia University — New York, NY
MS in Applied Mathematics, GPA: 3.9/4.00 — Aug 2025 – Dec 2026

Sapienza University of Rome — Rome, Italy
BS in Mathematical Sciences for AI, GPA: 3.99/4.00 | 110/110 *summa cum laude* — Sep 2022 – Sep 2025
Achievement: Earned two B.S. degrees with highest honors, pursuing both concurrently at separate universities.

University of Bologna — Bologna, Italy
BS in Statistical Sciences, GPA: 3.95/4.00 | 110/110 *summa cum laude* — Sep 2021 – Jul 2024
Award: University of Bologna Prize for Outstanding Student

Research Experience

Columbia University | Supervisor: Professor **Ioannis Karatzas** — New York, NY
Graduate Researcher — Dec 2025 – Present
Drift and Volatility Control with Discretionary Stopping – Working paper.

- Proved constant optimal drift and volatility controls on continuation tails for convex terminal costs; derived stopping thresholds and characterized free-boundary policies for nonconvex costs with multiple quadratic targets.
- Implemented a Python free-boundary optimizer validated by finite differences and Monte Carlo across 132 regimes; achieved **8%** median cost reduction vs. the best single-control baseline, with **0.37%** worst-case regret under ±10% parameter shifts.

Northeastern University | Supervisor: Professor **Francesco Restuccia** — Boston, MA
Statistical Theory Expert, ARO PECASE project on Credal Information Theory — Dec 2025 – Present
Credal Rate–Distortion Theory – *IEEE ITW 2026*; *extended version under review at IEEE Trans. Inf. Theory.*
N. Bonacorsi*, M. Caprio, F. Meneghello, F. Restuccia.*

- Formulated a robust rate–distortion theory for ML under model uncertainty; proved optimal one-shot coding reduces to incompatibility-graph coloring at fixed distortion and characterized asymptotic rates via graph entropy.

The Credal Information Bottleneck – submitted to ICLR 2027. *D.Loi, G.Onorato,* ***N.Bonacorsi****, F.Meneghello, F.Restuccia*

- Extended information bottleneck to credal uncertainty; across three benchmarks, reduced representation rate **35%** at matched worst-case accuracy and gained **6 pp** accuracy at matched rate vs. standard and robust IB baselines.

Work Experience

MENTIS Laboratory — Boston, MA
AI Research Intern — Jul 2026 – Aug 2026

- Engineered a Python combinatorial-optimization pipeline using LP feasibility tests and graph search; solved 54,000 problems, accelerated search **20**×, and stayed within 1% of the exact optimum.
- Built uncertainty-aware ML codebooks, compressing 60 credal sets into **14 codewords (4.29×)** with zero incompatibility violations versus **18.6%** for an equal-size barycentric baseline.

E.PTY Group — Bologna, Italy
Quantitative Researcher — Jun 2023 – Jun 2025

- Built a hierarchical Bayesian forecasting and Monte Carlo allocation engine; reduced rolling out-of-sample RMSE by **14.6%** versus historical forecasting baselines.
- Deployed downside-constrained stochastic allocation, increasing realized operating profit by **11.8%**; historical backtests showed **17.3%** lower 95% CVaR.

Quantitative Projects

Certified Alpha Capacity (AlphaValue) [arXiv] [Code] [Calculator] — Aug 2026 – Present

- Selected as **ML-Quant's #1 arXiv Top Pick [Newsletter]**.
- Developed and released *AlphaValue*, an open-source Python framework for statistical arbitrage signal certification; derived exact feasibility thresholds, sharp optimal-stopping capacity law, and competitive equilibrium for endogenous alpha decay.
- Validated *AlphaValue* on 543 BTC/ETH funding events: lifetime score outperformed current edge in seven-day funding rank correlation (**0.474 vs. 0.383**), retaining **0.2777** incremental rank association (95% CI **[0.123,0.406]**).

Maturity-Safe Distributional Volatility Forecasting [Report] [Code] — May 2026 – Jul 2026

- Assembled a leakage-safe 22-day variance forecaster using six experts, bounded-loss Hedge, maturity-delayed updates, and 22 non-overlapping residual streams.
- Evaluated 3,381 out-of-sample forecasts, reducing log-RMSE by **14.9% vs. VIX** (HAC $p < 0.001$) with **89%** coverage at nominal 90%; volatility targeting lowered max drawdown from **15.3% to 11.6%** versus fixed exposure.

Bootstrap-Calibrated Distributionally Robust Allocation [Report] [Code] — May 2026 – Jul 2026

- Designed a DRO allocator for 25 Fama–French portfolios using circular-block-bootstrap covariance uncertainty and capped long-only QPs with a custom active-set solver.
- Achieved **16.45%** OOS volatility and **0.820** net Sharpe after 10 bps, matching Ledoit–Wolf GMV while cutting volatility **18.4%** versus equal weight.

Skills

Programming & Data: Python, SQL, R, MATLAB; NumPy, pandas, SciPy, scikit-learn, PyTorch, statsmodels.
Methods: Probability, Bayesian Inference, Time Series, Monte Carlo, Machine Learning, Stochastic Processes, Stochastic Control, Optimal Stopping, Optimization, Backtesting, Risk Modeling, Portfolio Optimization.
Languages: Italian (native), English (fluent).